Machine Translation and Implementation of Transformers

Submitted in partial fulfilment of
the credit requirements for the degree of
Bachelor of Technology with Honours
in Computer Science and Engineering
(Artificial Intelligence and Machine Learning)

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Abstract

This report covers the topics presented in the "Introductions to Large Language Models" course taught by Tanmoy Chakraborty and Soumen Chakrabarti, available on NPTEL. As a graded assignment, the report follows a set of seven questions, and accompanies a code base available in the repository linked below. Each question will be quoted in full, followed by explanations and code listings where appropriate.

Note To Readers

A repository accompanying this report is available at the author's GitHub homepage. It is accessible to the public and can be navigated to by clicking or scanning the QR Code on the right.

The repository contains source code used for the development of models described in this report in the `code/` directory. The repository also contains the LaTeXsource code used for the compilation of this report in the `report/` directory. A `flake.nix` and `flake.lock` have also been provided for readers familiar with Nix to reproduce the development environment.

QR Code linking to accompanying repository

Work Attribution

All work for this project has been carried out without any AI assistance or code/text generation. A work history is provided at the end of the report as a proof of work. This work was solely developed by me, P Nakul Bhat, without any undeclared assistance. All sources used for references have been declared in the report, and are also available as a bibliography file in `report/` directory.

1 Sequence-to-Sequence Model

(a) Implement a Transformer-based sequence-to-sequence model in PyTorch for machine translation (English → French).

Implementation is available in the listings ...**??**

(b) Explain the encoder-decoder structure, attention mechanism, and training process.
The model is like this [1]

(c) Provide code snippets and discuss evaluation metrics such as BLEU score.

References

[1] Kyle Aitken et al. *Understanding How Encoder-Decoder Architectures Attend.* Oct. 28, 2021. DOI: `10.48550/arXiv.2110.15253`. arXiv: `2110.15253 [cs]`. URL: `http://arxiv.org/abs/2110.15253` (visited on 12/10/2025). Pre-published.